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APPARATUS AND METHODS FOR TREATING ROOT CANALS OF TEETH

Abstract

Apparatus and methods for endodontic treatment of teeth provide effective cleaning of organic material (such as pulp and diseased tissue) from the root canal system. In an embodiment, a compressor system generates high pressure liquid (e.g., water) that flows through an orifice to produce a high-velocity collimated jet of liquid. The high-velocity jet is directed toward a surface of a tooth, for example, an exposed dentinal surface, and impingement of the jet onto the surface generates an acoustic wave that propagates throughout the tooth. The acoustic wave effectively detaches organic material from dentinal surfaces and tubules. The detached organic material is flushed from the root canal system by the liquid jet and/or by additional irrigation.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/958,857, filed Oct. 3, 2022, entitled “APPARATUS AND METHODS FOR TREATING ROOT CANALS OF TEETH,” which is a continuation of U.S. patent application Ser. No. 16/951,968, filed Nov. 18, 2020, entitled “APPARATUS AND METHODS FOR TREATING ROOT CANALS OF TEETH,” which is a continuation of U.S. patent application Ser. No. 16/546,032, filed Aug. 20, 2019, entitled “APPARATUS AND METHODS FOR TREATING ROOT CANALS OF TEETH,” which is a continuation of U.S. patent application Ser. No. 16/412,919, filed May 15, 2019, entitled “APPARATUS AND METHODS FOR TREATING ROOT CANALS OF TEETH,” which is a continuation of U.S. patent application Ser. No. 16/366,954, filed Mar. 27, 2019, entitled “APPARATUS AND METHODS FOR TREATING ROOT CANALS OF TEETH,” which is a continuation of U.S. patent application Ser. No. 14/628,500, filed Feb. 23, 2015, entitled “APPARATUS AND METHODS FOR TREATING ROOT CANALS OF TEETH,” which is a continuation of U.S. patent application Ser. No. 14/304,737, filed Jun. 13, 2014, entitled “APPARATUS AND METHODS FOR TREATING ROOT CANALS OF TEETH,” which is a continuation of U.S. patent application Ser. No. 11/737,710, filed Apr. 19, 2007, entitled “APPARATUS AND METHODS FOR TREATING ROOT CANALS OF TEETH,” which claims the benefit under 35 U.S.C. § 119 (e) of U.S. Provisional Patent Application No. 60/793,452, filed Apr. 20, 2006, entitled “APPARATUS AND METHODS FOR TREATING ROOT CANALS OF TEETH,” the entire contents of each of which is hereby incorporated by reference herein.

BACKGROUND

Field of the Disclosure

[0002] The present disclosure generally relates to methods and apparatus for removing organic matter from a body location and, more particularly, to methods and apparatus for removing organic matter from a root canal system of a tooth.

Description of the Related Art

[0003] In conventional root canal procedures, an opening is drilled through the crown of a diseased tooth, and endodontic files are inserted into the root canal system to open the canal and remove organic material therein. The root canal is then filled with solid matter such as gutta percha, and the

tooth is restored. However, this procedure will not remove all organic material from all canal spaces. The action of the file during the process of opening the canal creates a smear layer of dentinal filings and diseased organic material on the dentinal walls, which is extremely difficult to remove. The organic material and necrotic tissue that remain in the canal spaces after completion of the procedure often result in post-procedure complications such as infections.

SUMMARY

[0004] In an embodiment, an apparatus for removing organic material from a root canal of a tooth is provided. The apparatus may comprise a liquid jet assembly having a liquid pressurization portion which pressurizes a liquid and a liquid beam forming portion in fluid communication with the pressurization portion. The beam forming portion may comprise an orifice that receives the pressurized liquid. The orifice may be sized and shaped to convert the pressurized liquid into a high velocity collimated beam that produces an acoustic wave upon impact with a surface of the tooth. The energy of the wave may cause organic material within the canal to be detached from the surrounding dentinal surface along a length of the canal. The length of the canal may extend at least to an apical portion of the tooth.

[0005] In another embodiment, a method of removing organic material that fills a root canal of a tooth is provided. The method comprises propagating an acoustic wave through the tooth. The method may also comprise detaching organic material filling the canal from the surrounding dentinal tissue using energy of the acoustic wave.

[0006] In another embodiment, a method of removing organic material from dentinal tubules which extend laterally from a root canal is provided. The method comprises introducing energy into a plurality of tubules through dentinal tissue such that at least a portion of an odontoblastic process within the tubules is detached from surrounding dentinal tissue and released from the tubule.

[0007] In another embodiment, a method for removing organic material from a root canal of a tooth is provided. The method comprises impacting dentin with an energy beam of a sufficiently high level to cause cavitations in fluid within the root canal. The cavitations maybe caused at least at locations in the root canal remote relative to the location of energy impact such that organic material within the canal may be detached from surrounding dentinal tissue.

[0008] In another embodiment, a method of removing organic material from a root canal comprises directing a liquid jet into the pulp chamber of a tooth through an opening in a side of the tooth at a substantial angle to the long axis of a root canal.

[0009] In another embodiment, a method of removing organic material from a root canal using a high velocity liquid jet is provided. The method comprises providing a handpiece for directing the liquid jet and positioning a contact member of the handpiece against a tooth to be treated. The method may also comprise using a sensor to sense contact of the contact member with the tooth. The method also may comprise activating the liquid jet only after the contact is sensed by the sensor.

[0010] In another embodiment, a method of removing organic material from a tooth is provided. The method comprises using acoustic energy to detach organic material from surrounding dentin within a plurality of root canals of a single tooth substantially simultaneously.

[0011] In another embodiment, an apparatus for removing organic material from a root canal of a tooth is provided. The apparatus comprises an acoustic energy generator arranged to couple acoustic energy to a dentinal surface of the tooth. The acoustic energy may be sufficient to cause organic material in the tooth to be detached from surrounding dentin at locations remote from the acoustic coupling surface.

[0012] In another embodiment, a method of removing organic material from a pulp cavity of a tooth is provided. The method comprises providing a liquid jet beam by passing liquid through an orifice. The method may also comprise using a positioning member to position the orifice relative to an opening into a pulp cavity of the tooth such that the jet beam passes through the opening.

[0013] For purposes of this summary, certain aspects, advantages, and novel features of the

invention are described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any particular embodiment of the invention. Thus, for example, those skilled in the art will recognize that the invention may be embodied or carried out in a manner that achieves one advantage or group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a cross-section view schematically illustrating a root canal system of a tooth.

[0015] FIG. 2 is a scanning electron microscope photograph of a dentinal surface within a canal system in a tooth and shows numerous dentinal tubules on the dentinal surface.

[0016] FIG. 3 is a block diagram schematically illustrating an embodiment of a compressor system adapted to produce a high-velocity liquid jet.

[0017] FIGS. 4 and 4A are cross-section views schematically illustrating an embodiment of a handpiece that can be used to maneuver the high-velocity liquid jet.

[0018] FIG. 5A is cross-section view schematically illustrating a distal end of an embodiment of a handpiece configured to deliver a high-velocity liquid jet.

[0019] FIG. 5B is a graph showing an example velocity profile of a coherent collimated jet.

[0020] FIG. 6A is a cross-section view schematically showing an endodontic method in which a high-velocity jet is directed toward dentin through an opening in the top of a tooth.

[0021] FIG. 6B is a cross-section view schematically showing another endodontic method in which the high-velocity jet is directed toward the dentin through an inlet opening in a side of the tooth and a relief opening in the top of the tooth is provided to reduce pressure buildup, if present, and to permit debridement.

[0022] FIG. 6C is a cross-section view schematically illustrating an embodiment of a positioning member prior to adherence to a tooth.

[0023] FIG. 6D is a cross-section view schematically illustrating an embodiment of a positioning member adhered to a side of a tooth and used to assist coupling the distal end of the handpiece to the side of the tooth so that the high-velocity jet may be directed through the inlet opening.

[0024] FIG. 7 schematically illustrates production of an acoustic wave caused by impingement of a high-velocity liquid jet onto a dentinal surface.

[0025] FIG. 8A is a cross-section view schematically illustrating cavitation bubbles formed near odontoblasts at the dentinal surface.

[0026] FIG. 8B schematically illustrates collapse of a cavitation bubble and formation of a cavitation jet near a dentinal surface.

[0027] FIGS. 9A-9C are scanning electron microscope photographs of dentinal surfaces following treatment of the tooth with the high-velocity jet; FIG. 9A shows dentinal tubules in an apical area of a mature tooth magnified 1000×; FIGS. 9B and 9C show dentin and dentinal tubules magnified 1000× in an inclusion area of a juvenile tooth (FIG. 9B) and in a medial area of a mature root (FIG. 9C). A bar at the top left of each photo indicates the linear scale (in microns) for each photograph.

[0028] FIGS. 10A-10H schematically illustrate embodiments of a cap that may be attached to a distal end of a handpiece and fitted onto a tooth. FIGS. 10A and 10C-10H are side views, and FIG. 10B is a partially exploded cross-section view.

[0029] FIG. 11 schematically illustrates a method for generating an acoustic wave using a piezoelectric transducer.

[0030] FIG. 12 is a cross-sectional side view of an embodiment of a guide tube.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0031] The present disclosure provides various apparatus and methods for dental treatments that

overcome possible disadvantages associated with conventional root canal treatments. In certain embodiments, endodontic treatment methods (e.g., root canal therapy) comprise directing a high-velocity liquid jet toward a tooth. Impact of the jet causes acoustic energy to propagate from a site of impact through the entire tooth, including the root canal system of the tooth. The acoustic energy is effective at detaching substantially all organic material in the root canal system from surrounding dentinal walls. In many embodiments, the detached organic material can be flushed from the root canal using low-velocity irrigation fluid. As used herein organic material (or organic matter) includes organic substances typically found in healthy or diseased root canal systems such as, for example, soft tissue, pulp, blood vessels, nerves, connective tissue, cellular matter, pus, and microorganisms, whether living, inflamed, infected, diseased, necrotic, or decomposed.

[0032] FIG. 1 is a cross section schematically illustrating a typical human tooth **10**, which comprises a crown **12** extending above the gum tissue **14** and at least one root **16** set into a socket (alveolus) within the jaw bone **18**. Although the tooth **10** schematically depicted in FIG. 1 is a molar, the apparatus and methods described herein may be used on any type of tooth such as an incisor, a canine, a bicuspid, or a molar. The hard tissue of the tooth **10** includes dentin **20** which provides the primary structure of the tooth **10**, a very hard enamel layer **22** which covers the crown **12** to a cementoenamel junction **15** near the gum **14**, and cementum **24** which covers the dentin **20** of the tooth **10** below the cementoenamel junction **15**.

[0033] A pulp cavity **26** is defined within the dentin **20**. The pulp cavity **26** comprises a pulp chamber **28** in the crown **11** and a root canal space **30** extending toward an apex **32** of each root **16**. The pulp cavity **26** contains dental pulp, which is a soft, vascular tissue comprising nerves, blood vessels, connective tissue, odontoblasts, and other tissue and cellular components. The pulp provides innervation and sustenance to the tooth through the epithelial lining of the pulp chamber **26** and the root canal space **30**. Blood vessels and nerves enter/exit the root canal space **30** through a tiny opening, the apical foramen **32**, near a tip of the apex **32** of the root **16**.

[0034] FIG. 2 depicts a pulpal surface of the dentin **20**. The dentin **20** comprises numerous, closely-packed, microscopic channels called dentinal tubules **34** that radiate outwards from the interior walls of the canal space **30** through the dentin **20** toward the exterior cementum **24** or enamel **22**. The tubules **34** run substantially parallel to each other and have diameters in a range from about 1.0 to 3.0 microns. The density of the tubules **34** is about 5,000-10,000 per mm.^{sup.2} near the apex **32** and increases to about 15,000 per mm.^{sup.2} near the crown.

[0035] The dentin **20** is continuously formed by specialized cells called odontoblasts that secrete a mineralized substance that hardens into dentin. Odontoblasts form a single layer of cells between the dentin **20** and the pulp. An odontoblast has a cell body that is located on the pulpal surface of the dentin **20** on a tubule **34** and a cytoplasmic portion, called the odontoblastic process, that extends into and substantially fills the associated tubule **34**. The odontoblasts are connected to each other with interodontoblastic collagen, and collagen fibrils may attach the odontoblast layer to the pulp. As a person ages, the odontoblasts continue to form dentin, which causes the root canal space **30** to decrease in diameter.

[0036] FIG. 3 is a block diagram that schematically illustrates a compressor system **38** adapted to generate a high-velocity jet of fluid for use in dental procedures. The compressor system **38** comprises a source of compressed gas **40** such as a pressurized air or gas source commonly available in dental service installations. The compressed gas **40** may be pressurized in a range from about 50 pounds per square inch (psi) to 150 psi including, for example, 100 psi. The compressed gas **40** may comprise any suitable commercially available gas including, for example, air, nitrogen, carbon dioxide, or a combination thereof. The compressed gas **40** is pneumatically connected to a pump **46** via a regulator **42**. The regulator **42** can be used to regulate the pressure of the input gas to a desired pressure such as, for example, 40 psi. In some embodiments, the pump **46** comprises an air-driven hydraulic pressure intensifier that uses the compressed gas **40** to increase the pressure of liquid received from a fluid source **44**. For example, a pressure intensifier having a 330:1 pressure

intensification ratio can increase the pressure of the liquid to about 13,200 psi using pressurized gas at 40 psi from the regulator **42**. Different pressure intensification ratios may be used in different embodiments. By adjusting the gas pressure with the regulator **42**, the pressure of the liquid output from the pump **46** can be selectably adjusted to a desired value or range of values. In some embodiments, the pressure of the liquid can be adjusted within a range from about 500 psi to about 50,000 psi. In certain embodiment, it has been found that a pressure range from about 2,000 psi to about 11,000 psi produces jets that are particularly effective for endodontic treatments.

[0037] The fluid source **44** may comprise a fluid container (e.g., an intravenous bag) holding sterile water, a medical-grade saline solution, an antiseptic or antibiotic solution, an abrasive solution, a solution with chemicals or medications, or any combination thereof. More than one fluid source may be used. In certain embodiments, it is advantageous for jet formation if the liquid provided by the fluid source **44** is substantially free of dissolved gases (e.g., less than 0.1% by volume) and particulates, which can act as nucleation sites for bubbles. In some embodiments, the fluid source **44** comprises degassed distilled water. A bubble detector (not shown) may be disposed between the fluid source **44** and the pump **46** to detect bubbles in the liquid and/or to determine whether liquid flow from the fluid source **44** has been interrupted or the container has emptied. The liquid in the fluid source **44** may be at room temperature or may be heated and/or cooled to a different temperature. For example, in some embodiments, the liquid in the fluid source **44** is chilled to reduce the temperature of the high velocity jet generated by the compressor system **38**.

[0038] In the embodiment depicted in FIG. **3**, the high-pressure liquid from the pump **46** is fed to a regulator **48** and then to a handpiece **50**, for example, by a length of high-pressure tubing **49**. The regulator **48** may be operable with compressed gas from the source **40** and may be used to regulate the pressure of the liquid to a desired value. For example, in one embodiment, the regulator **48** reduces the 13,200 psi pressure from the pump **46** to about 12,000 psi. The regulator **48** may include water pressure sensors and bleed-off valves (e.g., an air-driven needle valve) to provide the desired pressure and to permit an operator to actuate/deactuate water jet output from the handpiece **50**.

[0039] The handpiece **50** receives the high pressure liquid and is adapted at a distal end to generate a high-velocity, coherent, collimated beam or jet **60** of liquid for use in dental procedures. The handpiece **50** may be sized and shaped to be maneuverable so that the jet **60** may be directed toward or away from various portions of the tooth **10**.

[0040] The compressor system **38** may include a controller **51** that controls the operation of the components of the system **38**. The controller **51** may comprise a microprocessor, a special or general purpose computer, a floating point gate array, and/or a programmable logic device. In one embodiment, the controller **51** is used to operate the regulators **42**, **48** and the pump **46** so that the high-pressure liquid delivered to the handpiece **50** is at a suitable working pressure. The controller **51** may also be used to control safety of the system **38**, for example, by limiting system pressures to be below safety thresholds and/or by limiting the time that the jet **60** is permitted to flow from the handpiece **50**. In certain embodiments, the controller **51** may be used to vary or cycle the pressure of the liquid delivered to the handpiece **50**, for example, by cycling pressures provided by one or both of the regulators **42**, **48**. In certain such embodiments, sinusoidal or sawtooth pressure variability may be used to provide corresponding variability in the speed of the jet **60**. In certain embodiments, cycle time for the pressure variability may be in a range from about 0.1 seconds to about 5 seconds. Additionally and optionally, the controller **51** may regulate a pulse intensifier device (not shown), such as a piezoelectric transducer, that causes pulsations in the jet **60**. For example, in certain embodiments, the jet **60** comprises a pulsed jet, which may include a series of discrete liquid pulses, a continuous stream of fluid having spatially varying pressure, velocity, and/or area, or a combination thereof. The controller **51** advantageously may control the amplitude and frequency of the pulsations in the jet **60**. In certain embodiments, the amplitude of the pressure variation may be in a range from several hundred to several thousand psi. The pulse frequency may

be in a range from about 0.1 Hz to about 10 MHz. For example, in some embodiments, a pulse frequency of about 1 MHz is applied to produce a jet comprising a series of droplets.

[0041] The system **38** may also include a user interface **53** that outputs relevant system data and accepts user input. In some embodiments, the user interface **53** comprises a touch screen graphics display. In some embodiments, the user interface **53** may display information including the working liquid pressure in the handpiece **50** and instructions and/or procedures for operating on different tooth types (e.g., incisors, bicuspid, or molars). The user interface **53** may accept user input such as a time setting that sets a maximum time during which the compressor system **38** will deliver the jet **60** and other useful endodontic treatment options. For example, some embodiments permit an operator to select a “ramp up” and/or “ramp down” option in which the working liquid pressure can be gradually increased or decreased, respectively. The ramp up option advantageously may be used for initial aiming of the jet **60** towards a suitable portion of the tooth **10**, while the ramp down advantageously may be used if the jet **60** is moved toward a sensitive portion of the tooth **10** (e.g., the apex **32**). The compressor system **38** may also include a storage medium (e.g., volatile or nonvolatile memory) configured to store system operating information and executable instructions, user preferences, preferred operating pressures and times, patient data, etc. In some embodiments, the storage medium comprises on-board memory of the controller **51** and/or additional random access or read-only memory, flash memory, removable memory cards, etc.

[0042] The compressor system **38** may include additional and/or different components and may be configured differently than shown in FIG. 3. For example, the system **38** may include an aspiration pump that is coupled to the handpiece **50** (or an aspiration cannula) to permit aspiration of organic matter from the mouth or tooth **10**. In other embodiments, the compressor system **38** may comprise other pneumatic and/or hydraulic systems adapted to generate the high-velocity beam or jet **60**. For example, certain embodiments may utilize apparatus and systems described in U.S. Pat. No. 6,224,378, issued May 1, 2001, entitled “METHOD AND APPARATUS FOR DENTAL TREATMENT USING HIGH PRESSURE LIQUID JET,” and/or U.S. Pat. No. 6,497,572, issued Dec. 24, 2002, entitled “APPARATUS FOR DENTAL TREATMENT USING HIGH PRESSURE LIQUID JET,” the entire disclosure of each of which is hereby incorporated by reference herein.

[0043] Moreover, in other embodiments, the high-velocity jet **60** may be generated by systems other than the high-pressure compressor system **38**, such as, for example, by a pump system. In one such embodiment, an electric motor drives a pump that is in fluid communication with a liquid reservoir. The pump increases the velocity of the liquid so as to provide a narrow beam of high-velocity liquid from the handpiece **50**. In some embodiments, multiple pumps are used. As is well known from Bernoulli's law, the total pressure in a flowing fluid includes static (e.g., thermodynamic) pressure plus dynamic pressure (associated with fluid kinetic energy). A skilled artisan will recognize that static pressures in motor-driven pump systems may be less than static pressures in compressor systems, because the motor-driven pump primarily increases the dynamic pressure (e.g., the fluid velocity) of the liquid. The total pressures (static plus dynamic) achieved are comparable in many embodiments of compressor systems and pump systems.

[0044] FIGS. 4 and 4A are cross-section views that schematically illustrate one embodiment of the handpiece **50** adapted for forming the high-velocity jet **60**. The handpiece **50** comprises an elongated tubular barrel **52** having a central passageway **54** extending axially therethrough. The handpiece **50** has a proximal end **56** that is adapted to engage tubing from the regulator **48** in order for the passageway **54** to be in fluid communication with the high pressure liquid delivered by the compressor system **38**. The barrel **52** may include features **55** that enhance grasping the handpiece with the fingers and thumb of the operator. A distal end **58** of the barrel **52** (shown in closeup in FIG. 4A) includes a threaded recess adapted to engage complementary threads of an orifice mount **62**, which is configured to hold an orifice jewel **64** at an end thereof. The orifice mount **62** is tightly screwed into the distal end **58** of the barrel **52** to secure the orifice jewel **64** adjacent to a distal end of the passageway **52**.

[0045] The orifice jewel **64** may comprise a circular, disc-like element having a small, substantially central orifice **66** formed therein. The orifice jewel **64** may be fabricated from a suitably rigid material that resists deformation under high pressure such as, for example, synthetic sapphire or ruby. The orifice mount **62** advantageously secures the orifice jewel **64** substantially perpendicular to the passageway **54** so that high pressure liquid in the passageway **54** can flow through the orifice **66** and emerge as a highly collimated beam of fluid traveling along a longitudinal jet axis **70** that is substantially coaxial with the barrel **52** of the handpiece **50**. In some embodiments, the distal end **58** of the handpiece **50** may include additional components, for example, to assist guiding or directing the jet **60** and/or to provide aspiration. Also, as further described below, the distal end **58** of the handpiece **50** may be adapted to receive various end caps that assist guiding the jet **60** toward the pulp cavity **26**.

[0046] FIG. 5A is a cross-section schematically illustrating the distal end **58** of an embodiment of the handpiece **50** to further illustrate formation of the jet **60**. The orifice jewel **64** is secured at the distal end of the handpiece **50** and forms a tight seal to prevent leakage of high-pressure liquid **68** contained in the passageway **54**. In the depicted embodiment, the orifice jewel **64** has a proximal side that is substantially flat and a distal side that is concave (e.g., thinnest near the orifice **66**). The orifice **66** has a substantially circular cross-section with a diameter “D.” The axial length of sides **69** of the orifice **66** is “L.” The diameter D may be in a range from about 5 microns to about 1000 microns. Other diameter ranges are possible. In various embodiments, the diameter D may be in a range from about 10 microns to about 100 microns, a range from about 100 microns to about 500 microns, or range from about 500 microns to about 1000 microns. In various preferred embodiments, the orifice diameter D may be in a range of about 40-80 microns, a range of about 45-70 microns, or a range of about 45-65 microns. In one embodiment, the orifice diameter D is about 60 microns. The ratio of axial length L to diameter D may be about 50:1, 20:1, 10:1, 5:1, 1:1, or less. In one embodiment, the axial length L is about 500 microns. In certain embodiments, the ratio of axial length L to diameter D is selected so that transverse width of any boundary layers that may form on the sides **69** of the orifice **66** have a transverse width that is sufficiently small, for example, much less than the diameter D. In preferred embodiments, the orifice diameter is 40-80 microns, and more preferably 45-70 microns, and even more preferably 45-65 microns. The axial length of the orifice is preferably no greater than ten times the diameter of the orifice, and the liquid pressure at the input side of the orifice is 7,000 to 15,000 psi. In one embodiment, the orifice is about 60 microns in diameter, the axial length of the orifice about 500 microns, and the liquid pressure at the input side of the orifice about 11,000 psi. Since the output side of the orifice is at atmospheric pressure, the pressure drop at the orifice will be only slightly less than the pressure at the input side of the orifice. This combination of parameters provides a high velocity, high momentum, collimated, coherent liquid beam that is efficacious for cleaning without significant dentinal erosion.

[0047] In certain embodiments, the sides **69** of the orifice **66** are machined, polished, or otherwise processed to be substantially smooth in order to reduce or prevent formation of turbulence, cavitation, bubbles, fluid instabilities, or other effects that may interfere with substantially smooth, laminar flow of the liquid through the orifice **66**. For example, in certain such embodiments, the sides **69** have a root-mean-square (rms) surface roughness less than about 10 microns, less than about 1 micron, or less than about 0.1 microns. In other embodiments, the rms surface roughness is much smaller than the diameter D of the orifice such as, for example, less than about 0.1 D, less than about 0.01 D, less than about 0.001 D, or less than about 0.0001 D. Additionally, highly demineralized liquids may be used to reduce buildup of impurities along the sides **69**, which advantageously may increase the useful operating lifetime of the orifice jewel **64**.

[0048] As schematically depicted in FIG. 5A, the high pressure liquid **68** in the passageway **54** emerges through the orifice **66** as a high-velocity, collimated jet **60** traveling substantially along the jet axis **70** with a velocity, v. In some embodiments of the compressor system **38**, the jet velocity is

estimated to be proportional to $(P/\rho)^{1/2}$, where P is the liquid pressure in the passageway **54** and ρ is the density of the liquid. In certain embodiments, water pressurized to about 10,700 psi emerges from the orifice as a jet **60** having a velocity of about 220 m/s. By adjusting the liquid pressure delivered by the compressor system **38**, the handpiece **50** can deliver jets having different velocities. In some embodiments, the user interface **53** permits the operator to selectively adjust system pressures so that the velocity of the jet is suitable for a particular dental treatment.

[0049] In certain embodiments of the system **38**, the liquid used to form the jet **60** is substantially free from dissolved gases (e.g., less than about 0.1% per volume). If the dissolved gas content of the liquid is too high, bubbles may be formed at the nozzle orifice **66** due to the pressure drop.

Additionally, the pressure drop should preferably be sufficiently low to prevent formation of vapor at the distal end of the orifice **66**. The presence of substantial vapor, gas or bubbles, or particle contaminants in the liquid may cause a significant portion of the energy of the liquid jet **60** to be depleted, and there may be insufficient kinetic energy (and/or momentum) to provide efficient cleaning of the root canal system. When used for removing tissue and/or organic matter from root canals, the effectiveness of the device disclosed in U.S. Pat. No. 6,497,572, issued Dec. 24, 2002, and entitled "Apparatus for Dental Treatment Using High-Pressure Liquid Jet," is significantly increased by using liquids that are free (or at least substantially free) of dissolved gases (as well as bubbles) to form the high-velocity jet. Preferably, the liquid is bubble-free distilled water, and the concentration of dissolved gases is no more than e.g. 0.1% by volume. In use, the liquid beam is preferably directed at the floor of the pulp chamber at an oblique angle relative to the long axis of the root canals. Although the chamber fills with liquid, the beam has sufficient velocity to impact the submerged dentin with great force. Upon impingement, the primary, collimated coherent beam from the jet apparatus generates a high-energy acoustic pressure wave that propagates along the body of the tooth. At the dentinal surfaces of the main and side canals, the acoustic wave causes any surrounding liquid to cavitate. This cavitation is a surface effect cavitation caused by conversion of the water (or other liquid) from a liquid state to a vapor state. Due to the high energy required for such conversion, collapse of the cavitation bubble occurs with great force against the surface of the dentin and cleans through creation of cavitation-induced sub-jets which radiate inward toward the surface from the point of collapse of the cavitating vapor. The substantially gas-free liquid is preferred for the above described cavitation process. If the dissolved gas content of the liquid is too high, bubbles will be formed at the nozzle orifice due to the pressure drop.

Additionally, the pressure drop should preferably be sufficiently low to prevent formation of vapor at the nozzle orifice. The presence of significant vapor, gas or bubbles causes much of the energy of the beam to be depleted, and there will be insufficient energy to generate the liquid to vapor-phase cavitation, which is a surface effect.

[0050] The jet **60** emerges from the distal end of the orifice **66** as a beam of fluid traveling substantially parallel to the jet axis **70**. Such jets are called "collimated" jets. In certain embodiments, the angular divergence of the jet **60** is less than about 1 degree, less than about 0.1 degree, or less than about 0.01 degree. In other embodiments, jet beams with different angular divergences may be used. In some embodiments, the jet **60** can travel as a collimated beam for a distance of about 1 to 3 inches before the jet **60** begins to disperse (e.g., due to entrainment of air). In certain embodiments, the jet **60** may travel a distance at least several thousand times the jet diameter D before beginning to disperse.

[0051] As described above, it may be advantageous for the sides **69** of the orifice **66** to be sufficiently smooth that liquid flows through the orifice **66** in a substantially laminar manner. In certain embodiments, the transverse width of any boundary layers formed along the sides **69** of the orifice **66** is much smaller than the diameter D , and, away from the boundary layers, the speed of the jet **60** is substantially constant across the width of the orifice. FIG. 5B is a graph schematically illustrating an example velocity profile of the jet **60** after it has emerged from the distal end of the orifice **66**. The graph depicts flow velocity in the direction of the jet axis **70** versus a distance

transverse (e.g., orthogonal) to the jet axis **70**. In this example embodiment, away from narrow boundary layers near the outer surface of the jet **60** (e.g., near 0 and D on the graph), the jet velocity is substantially constant across the width of the jet. Jets having substantially constant velocity profiles are called “coherent” jets. In other embodiments, the velocity profile of the jet **60** is not substantially constant across the width of the jet **60**. For example, the jet velocity profile in certain embodiments is a parabolic profile well-known from pipe flow.

[0052] In certain embodiments, the compressor system **38** is configured to deliver a coherent, collimated jet **60** of high-velocity liquid. A coherent, collimated jet will be denoted herein as a “CC jet.” The following example provides various representative properties of a CC jet **60** that can be generated using an embodiment of the system **38**. In this example system, the diameter D and the axial length L of the orifice **66** are 60 microns and 500 microns, respectively. In one embodiment, the pressure of the liquid (degassed, distilled water) in the handpiece **50** is about 8,000 psi, which produces a jet velocity of about 190 m/s. The mass discharge rate of the jet is about 0.5 g/s, and the jet can produce a force of about 0.1 Newton when impacting a surface at normal incidence. The jet provides a kinetic power of about 10 Watts. If the jet is directed toward a tooth for about 10 seconds, the jet can deliver a momentum (or impulse) of about 1 kg m/s and an energy of about 100 Joules (about 23 calories).

[0053] In other embodiments, the CC jet is produced from liquid pressurized to about 2500 psi. The jet velocity is about 110 m/s, and volume flow rate is about 0.3 mL/s. The CC jet can produce about 2 W of kinetic power. The jet **60** may remain substantially collimated over propagation lengths from about 1 cm to about 30 cm in various embodiments.

[0054] The energy flux produced by the liquid jet is the kinetic power of the jet divided by the transverse area of the jet beam. The energy flux may be in a range from about 1 kW/cm.² to about 1000 kW/cm.². In some embodiments, the energy flux is in a range from about 50 kW/cm.² to about 750 kW/cm.², including, for example, 70 kW/cm.², 175 kW/cm.², 350 kW/cm.², and 550 kW/cm.². In one experiment, a CC jet was directed toward a dentinal surface of a tooth, and widespread acoustic noise (possibly due to acoustic cavitation) was detected in the tooth when the energy flux of the jet exceeded about 75 kW/cm.². At the onset of detectable acoustic noise, the CC jet had the following properties: velocity of about 110 m/s, kinetic power of about 2 W, and mass flow rate of about 0.3 g/s. The pressure producing the CC jet was about 2500 psi.

[0055] By using different fluid working pressures and/or orifice diameters, jets having different properties can be generated. For example, in certain embodiments, the mass discharge rate may be in a range from about 0.01 g/s to about 1 g/s, the jet velocity may be in range from about 50 m/s to about 300 m/s, the jet force may be in a range from about 0.01 N to about 1 N, and the jet power may be in a range from about 0.1 W to about 50 W. In various endodontic treatments, the jet is applied to a tooth for a time in a range from about 1 second to 120 seconds. Accordingly, in such treatments, the jet can deliver momentum (or impulse) in a range of about 0.01 kg m/s to about 100 kg m/s, and energy in a range of about 0.1 J to about 500 J. In some embodiments, an energy range from about 20 J to about 400 J may be effective at providing cleaning of the root canal system without causing substantial erosion of dentin. A person of ordinary skill will recognize that the compressor system **38** can be configured to provide liquid jets having a wide range of properties that may be different from the example values and ranges provided herein, which are intended to be illustrative and non-limiting.

[0056] In various dental treatments, the compressor system **38** delivers a jet **60**, which advantageously may be a CC jet, that is directed toward one or more portions of a tooth in order to, for example, excise and/or emulsify organic material, provide irrigation, and/or generate acoustic energy for delaminating organic matter from the pulp cavity **26**.

[0057] FIG. **6A** schematically illustrates one embodiment of an endodontic treatment for diseased pulp in the tooth **10**. A drill or grinding tool is initially used to make an opening **80** in the tooth **10**.

The opening **80** may extend through the enamel **22** and the dentin **20** to expose and provide access to pulp in the pulp cavity **26**. The opening **80** may be made in a top portion of the crown **12** of the tooth **10** (as shown in FIG. 3) or in another portion such as a side of the crown **12** or in the root **16** below the gum **14**. The opening **80** may be sized and shaped as needed to provide suitable access to the diseased pulp and/or some or all of the canal spaces **30**. In some treatment methods, additional openings may be formed in the tooth **10** to provide further access to the pulp and/or to provide dental irrigation.

[0058] The handpiece **50** may be used to deliver a jet **60** of liquid to a portion of the tooth **10**. The jet **60** advantageously may, but need not, be a CC jet. The jet **60** can be used to cut through organic material in the pulp chamber **28**. Additionally, as will be further described below, the jet **60** may be directed toward hard surfaces of the tooth **10** (e.g., the dentin **20**) to generate acoustic energy, which may propagate through the dentin **20**, the dentinal tubule, and the organic material in the root canal space **30**. The acoustic energy causes detachment of organic material from the dentin **20** without requiring that the jet directly impact the organic material. In certain embodiments, the acoustic energy has been found to be effective in causing detachment of the entire body of pulp (and other organic material) from within the pulp chamber **28** and/or root canal space **30**, without the use of endodontic files. The jet **60** preferably should have insufficient energy, energy flux, and/or momentum to damage or substantially erode the dentin **20**.

[0059] In some treatment methods, the operator can maneuver the handpiece **50** to direct the jet **60** around the pulp chamber **28** during the treatment process. The distal end of the handpiece **50** may be held about 1 inch from the tooth **10** so that the liquid impacts a portion of the tooth **10** as a substantially collimated coherent beam. The jet **60** may provide sufficient force to cut through and/or emulsify some or all of the organic material in the pulp chamber **28**. The flow of liquid from the jet **60** may create sufficient swirling or turbulent flow to remove the cut and/or emulsified organic material from the pulp cavity **26** so that it can be aspirated from the mouth of the patient. In other treatment embodiments, pulpal tissue in the pulp chamber **28** may be removed via conventional techniques prior to (and/or during) liquid jet treatment to expose a portion of the dentin **20**. The jet **60** may then be directed to the exposed portion.

[0060] The jet **60** may be directed toward the floor **82** of the pulp chamber **28** (see, e.g., FIG. 7). In some methods, the jet **60** is directed toward the floor **82** (and/or walls) of the pulp chamber **28** advantageously at a substantial angle (e.g., 15-50 degrees) relative to the long axis of the root canal space **30** to ensure that the jet does not directly impact the apical portion of the canal space **30**, thereby reducing a possibility that the force (and/or pressure) imparted by the jet **60** will cause damage to healthy tissue around the apical foramen **34**. Accordingly, certain disclosed treatment methods advantageously may be used on “open apex” teeth having underdeveloped and/or enlarged apices, because impingement of the jet **60** in the pulp chamber **28** will not harm the periapical portion of the tooth **10**. Additionally or optionally, the jet **60** can be directed toward one or more sides of the pulp chamber **28** so as to impact the dentin **20**. In some embodiments, the jet is directed to several locations in or on the tooth **10**. An advantage of some methods is that the impact of the jet **60** on the dentin **20** does not cause significant erosion or destruction of the dentin **20** within the tooth **10**. Accordingly, such methods may be minimally invasive in comparison with conventional root canal procedures.

[0061] The pulp cavity **26** may fill with fluid during the treatment. For sufficiently high working pressures, the jet **60** will have sufficient velocity to impact submerged dentin **20** with enough force to provide effective treatment. In certain embodiments of the treatment method, one or more properties of the jet **60** are temporally varied during the treatment. For example, the working pressure may be varied to ramp up or ramp down the jet velocity or to cause the jet to alternate between high-speed flow and low-speed flow. The jet **60** may comprise a pulsed jet with pulsation amplitude and/or frequency selected to provide effective treatment. A combination of the above treatment methods may be used, serially or in alternation.

[0062] As noted above, detachment of the organic material within the root canal system from the surrounding dentin **20** does not require that the jet **60** impact the organic material in the root canal space **30**. Rather, the jet **60** may be directed against a dentinal wall (e.g., in the pulp chamber **28**), which couples acoustic energy to the tooth **10** so as to cause detachment of the organic material. In some methods, the detachment occurs relatively quickly after the jet **60** impinges on the dentinal wall (e.g., within a few seconds) and may occur substantially simultaneously throughout one or more root canal spaces **30** of the tooth **10**.

[0063] In one presently preferred method schematically illustrated in FIG. **6B**, the jet beam **60** is introduced into an inlet opening **84** formed in the side (e.g., buccal or lingual surface) of the tooth **10** with a conventional dental drill. The opening **84** may be formed near the cementoenamel junction **15**. In some procedures, a portion of the gum **14** is depressed to provide access to the intended position of the opening **84** near the cementoenamel junction **15**. The opening **84** may have a diameter in a range from about 1 mm to about 2 mm and may extend from the exterior surface of the tooth **10** to the pulp chamber **28**. In some embodiments, the diameter is about 1.2 mm. Different diameters may be used in other embodiments. The opening **84** is thus generally transverse to the long axis of any root canal space **30** and ensures that the energy of the jet **60** will not be directed down any canal spaces **30**. A benefit of providing the opening **84** in the side of the tooth **10** is that less hard tissue of the tooth **10** is damaged than if the opening were formed through the occlusal surface of the tooth **10**. Although it is presently preferred to use a single inlet opening **84** to reduce invasiveness of the procedure, in other methods, two or more inlet openings may be used.

[0064] FIGS. **6C** and **6D** are cross-section views schematically illustrating an embodiment of a positioning member **130** that may be used to assist coupling and orienting the distal end **58** of the handpiece **50** to the tooth **10** so that the high-velocity jet **60** is directed through the inlet opening **84** in the side of the tooth **10**. In the illustrated embodiment, the positioning member **130** comprises a collar portion **134** that may be generally disk-like in shape. The collar portion **134** may have a width in a range from about 1 mm to about 10 mm. The collar portion **134** may have a substantially central opening **136** having a diameter approximately equal to the diameter of the inlet opening **84**. In some embodiments, the collar portion **134** is formed from a flexible material (such as an elastomer) so that it can conform to the surface of the tooth **10**. An adhesive (such as a light-cured orthodontic adhesive) may be included on a surface **134a** of the collar portion **134** to enable the positioning member **130** to adhere to the tooth **10**. In some embodiments, a detachable, elongated peg **138** may be used to position the position member **130** so that the central opening **136** in the collar portion **134** is aligned with the inlet opening **84** in the tooth **10**. A distal end of the peg **138** may be sized to fit within the opening **84**. When the positioning member **130** is in position on the tooth **10** and the adhesive has sufficiently cured, the peg **138** may be removed, leaving the positioning member **130** adhered to the side of the tooth **10** (see FIG. **6D**). Additionally or alternatively, the collar portion **134** may include alignment guides disposed near the opening **136** to assist positioning the member **130** over the opening **84**. For example, a circular ridge, having an outside diameter slightly smaller than the inside diameter of the opening **84**, may be formed on the surface **134a** around the opening **136** and used to align the openings **84** and **136**.

[0065] The positioning member **130** may include mounting portions **132a** configured to engage complementary mounting portions **132b** disposed on the distal end **58** of the handpiece **50**. For example, the mounting portions **132a**, **132b** may comprise a standard, quick-turn connector such as a Luer-lock. FIG. **6D** schematically illustrates the handpiece **50** before (or after) engagement with the positioning member **130**. When engaged with the positioning member **130**, the handpiece **50** advantageously is oriented so that the jet axis **70** is substantially longitudinally aligned with the inlet opening **84**. Moreover, the positioning member **130** may stabilize the handpiece **50** against unwanted movement. Accordingly, upon actuation, the jet **60** will be directed through the opening **84** and into the pulp cavity **26**. As further described below with reference to FIG. **10B**, the distal end **58** of the handpiece **50** may comprise one or more pressure sensors adapted to sense when the

distal end **58** is securely engaged with the positioning member **130**. In such embodiments, the system may not permit the jet **60** to be actuated until a sufficiently secure fit and proper alignment are indicated by signals from the pressure sensors.

[0066] When the distal end **58** of the handpiece **50** is engaged with the positioning member **130**, the jet **60** may be directed through the inlet opening **84** so that it impacts the dentinal wall and causes detachment of the organic material in the root canal spaces **30**. After the treatment is completed, the positioning member **130** may be removed from the tooth **10** using any well-known technique for releasing the adhesive. Remaining adhesive, if present, may function as bonding for restorative material used to close the defect.

[0067] To reduce possible buildup of fluid pressure within the pulp cavity **26**, a relief opening **88** may be formed in the top side (e.g., occlusal surface) of the tooth **10**. The relief opening **88** may be formed on a buccal or lingual surface. The diameter of the relief outlet or opening **88** may be larger than that of the inlet opening **84**, for example, about 2 mm to about 3 mm. In some methods, the diameter of the relief opening **88** may be about the same as (or smaller than) the diameter of the inlet opening **84** (e.g., about 1.2 mm in one embodiment). The relief opening **88** also serves to facilitate debridement and evacuation of the detached organic material. The diameter of the relief opening **88** advantageously may be large enough to permit flushing out pulp fragments. In some methods, two or more relief openings **88** may be used.

[0068] Those skilled in dentistry will recognize that the inlet and relief openings **84**, **88** are quite small relative to the openings required for conventional root canal procedures, thus preserving valuable tooth structure. For example, in some methods, even though two openings **84**, **88** are used, less tooth material is removed than in a conventional root canal procedure using a single, standard-sized occlusal opening. Moreover, many patients have existing coronal defects (e.g., decay, restorations, preparations, etc.), and it may be possible to form one or both of the openings simply by removing material other than healthy tooth tissue, such as a filling. Fillings on one or more sides of the tooth **10** may also be used to form the openings **84**, **88**. In any event, once the jet has caused acoustically induced detachment of the organic material, low pressure flushing fluid (such as water) may be introduced into either or both the openings **84**, **88** to irrigate the canal space **30** and flush out the organic material. Additionally and optionally, manual extraction of organic material may be performed with a dental instrument.

[0069] Certain teeth, particularly molars and/or wisdom teeth, may be difficult to access in conventional root canal therapies due to limited working space in the back of the mouth. Due to the difficulties or inconvenience of coronal access to these teeth in conventional root canal therapies, some of these teeth, which would otherwise be treatable, may instead be extracted. An advantage of some embodiments of the disclosed methods is that by permitting a wider range of access with the liquid jet **60** (e.g., on or through coronal, lingual, and/or buccal surfaces), the acoustic-induced detachment of organic material can save the tooth and reduce the likelihood of its extraction.

[0070] Without subscribing to any particular theory of operation, FIG. 7 schematically illustrates an explanation for the effectiveness of the treatment methods described herein. FIG. 7 depicts the jet **60** impacting the dentin **20** of the tooth **10**. Upon impact, a portion of the energy and/or momentum carried by the jet **60** generates an acoustic pressure wave **100** that propagates through the body of the tooth **10**. In addition to propagating through the dentin **20**, the acoustic wave **100** may propagate through organic material in the root canal space **30** and in the tubules of the dentin **20**. The acoustic wave **100** may include acoustic energy with acoustic frequencies in a range from about 1 Hz to above 5 MHz such as, for example, up to about 10 MHz. The acoustic wave **100** may have frequency components in the ultrasonic frequency range, e.g., above about 20 kHz. In some cases, the frequency range may include megasonic frequencies above about 1 MHz. The acoustic wave **100** may include other frequencies as well.

[0071] At the dentinal surfaces of the root canal space **30** and tubules, the acoustic wave **100** may cause surrounding liquid to cavitate. This cavitation may be a surface effect cavitation caused by

conversion of the water (or other liquid) from a liquid state to a vapor state. If the acoustic energy in the wave **100** is sufficiently large, the cavitation processes may include inertial cavitation wherein sufficiently low pressures caused by the acoustic wave **100** induce formation and collapse of bubbles in liquid near the dentinal surfaces. For smaller acoustic energies, non-inertial (or gas) cavitation may play a more significant role. In such cases, dissolved gases, tissue debris, and impurities act as nucleation centers for the formation of cavitation bubbles. Cavitation may also occur at pore sites across the microporous surface of the dentin **20**. The cavitation bubbles oscillate in response to the acoustic wave **100**, and amplitude of the oscillations may grow as additional gas is absorbed by the bubble.

[0072] Due to the relatively high energy required for formation of cavitation bubbles, collapse of the cavitation bubble occurs with great force against the surface of the dentin **20**. Bubble collapse near a surface is known to occur asymmetrically and may result in formation of cavitation jets that radiate toward the surface and produce locally very high pressures and/or elevated temperatures. In some cases, the acoustic wave **100** may also generate fluid motions (acoustic streaming) that enhance disruption of organic matter. Acoustic streaming also may be effective at transporting or flushing detached organic matter out of the root canal space **30** and/or the tubules.

[0073] Accordingly, in certain methods, the acoustic wave **100** cleans the root canal system through processes including formation and collapse of cavitation bubbles, radiation of cavitation jets toward dentinal surfaces, acoustic streaming, or a combination thereof. In the process, the organic material may be broken into small pieces or particles, which can be irrigated from the pulp cavity **26**. In some treatment methods, these cavitation processes may produce transient, localized high pressure and/or elevated temperature zones that disrupt, detach, and/or delaminate organic matter near root canal surfaces. For example, cavitation-induced effects may detach odontoblasts from the dentinal surface and effectively remove a portion of the odontoblastic process from the tubule. Cavitation-induced effects may also disrupt and/or detach the collagen fibrils that attach the odontoblast layer to the pulp in the interior of the canal space **30**. Cavitation-induced effects may also occur in interior regions of the pulp cavity **26** (e.g., away from the dentinal surfaces) and may disrupt and/or loosen organic material in the interior regions, thereby making this material more readily removable from the pulp cavity **26**.

[0074] Cavitation effects are believed to be formed everywhere the acoustic wave **100** propagates with sufficient energy. Accordingly, it is advantageous for the jet **60** to have sufficient energy and/or momentum to generate an acoustic wave **100** capable of causing cavitation effects throughout substantially the entire root canal system but without causing harm to the tooth **10**. For example, if the jet diameter D is too small and the momentum of the jet beam too high, impact of the beam may cause significant dentinal erosion. On the other hand, if the beam diameter D is too large and the momentum of the jet beam too low, the beam may have insufficient energy to produce an acoustic pressure wave **100** capable of causing cavitation. For example, in certain methods, the jet energy incident on the tooth is greater than about 20 J to provide effective cleaning but less than about 400 J to prevent dentinal erosion. In some methods, acoustic cavitation effects occur substantially throughout the root canal system when certain jet **60** properties are above threshold values. For example, in one experiment mentioned above, widespread acoustic noise (possibly caused by acoustic cavitation) was detected in a tooth only when the energy flux of the jet **60** was greater than about 75 kW/cm^{sup.2}. At the onset of detectable acoustic noise, the jet had a power of about 2 W, a velocity of about 115 m/s, a mass flow rate of about 0.3 g/s, and provided a force of about 0.03 N. The efficiency of conversion of jet kinetic energy into acoustic energy was estimated to be about 2%.

[0075] A portion of the acoustic wave **100** may also propagate through tissue and bone adjacent the tooth **10**. However, the energy carried by this portion of the acoustic wave **100** may be relatively small due to the acoustic impedance mismatch between the dentin **20** and nearby tissues and bone. Accordingly, cavitation-induced effects may be substantially reduced in tissues surrounding the

tooth **10**, and the endodontic methods described herein will not significantly damage the surrounding tissues.

[0076] FIGS. **8A** and **8B** are cross-section views that schematically illustrate some of the cavitation processes that clean a surface **104** of the dentin **20**. As depicted in FIG. **8A**, odontoblasts **106** are located near the surface and the odontoblastic process extends into the tubules **108**. The acoustic wave **100** induces oscillation and collapse of cavitation bubbles **110**, which generates transient localized heat and/or pressure that disrupts and detaches the odontoblasts **106** from the dentinal surface **104**. Cavitation bubbles may also form and collapse within the tubules **108**, thereby causing disruption of the odontoblastic processes in the tubules **108**. FIG. **8B** schematically depicts collapse of an initially spherical cavitation bubble **110** (shown in (i)) located near the body of organic matter **112** adjacent the dentinal surface **104** and filling the canal. In (ii), the side of the bubble **110** away from the surface **104** is perturbed from its spherical shape. In (iii), fluid **114** from the interior of the pulp cavity **26** penetrates the perturbed side of the bubble **110**. In (iv) the fluid **114** has formed a cavitation jet **116** radiating toward the surface **104**. The energy and momentum of the cavitation jet **116** breaks up and disperses the organic matter **112**.

[0077] FIGS. **9A-9C** are scanning electron microscope photographs of the dentinal surface showing the surprising effectiveness of the removal of organic matter by the acoustic effect. FIG. **9A** shows dentinal tubules in an apical area of a mature tooth magnified **1000x**. FIGS. **9B** and **9C** show dentin and dentinal tubules magnified **1000x** in an inclusion area of a juvenile tooth (FIG. **9B**) and in a medial area of a mature root (FIG. **9C**). A bar at the top left of each photo indicates the linear scale (in microns) for each photograph. As can be seen in FIGS. **9A-9C**, the dentinal surfaces are almost entirely free from organic matter, which appears to have been literally ripped away from the dentin. Flow of liquid in the root canal space **30** flushes and irrigates the organic matter from substantially all the root canal space **30** and the tubules. Returning to FIG. **2**, this photograph shows an apical area of a mature tooth magnified **2000x** and viewed at a slight slant from perpendicular to show that the tubules have been cleaned down to a distance of about 3 microns. FIGS. **2** and **9A-9C** demonstrate that cleaning of the dentinal surface is very effective and that almost no remnants of organic material remain after treatment.

[0078] As mentioned previously, it has been found that the cleaning does not require that the liquid jet **60** be aimed down the root canal space **30**, although that may be beneficial in certain isolated cases where the canal space **30** is very narrow and/or filled with dry material. Additionally, it has been found in some embodiments that the pulp cavity **26** need not be prepared or pre-treated (e.g., by removing root canal matter with one or more endodontic files) before application of the jet. Impingement of the jet **60** onto the dentin **20** in the pulp chamber **28** is sufficient in most cases to generate the acoustic wave **100** that causes the cleaning. Accordingly, in most cases, it is the acoustic wave **100** and not direct impact of the jet **60** that causes the cleaning, particularly for the dentinal surfaces near the apex of the root canal space **30**, which are remote from the pulp chamber **28**. For example, examination of FIGS. **2** and **9A-9C** shows that organic material has been removed from apical dentinal tubules, which are not possible to reach directly with the liquid jet beam. In certain embodiments, the jet beam **60** is capable of delivering sufficient energy to the tooth **10** to remove at least 90 percent of the organic material from the root canal system. In other embodiments, at least 95 percent of the organic material is removed. Preferably, the jet beam should be capable of removing substantially all the organic material from the root canal space **30** and from at least a portion of the tubules. The treatment time during which the high-velocity jet is directed toward the tooth **10** may range from about 1 second to about 120 seconds. In some embodiments, the treatment time is from about 10 seconds to about 30 seconds. In other embodiments, the treatment time is no more than 10 seconds such as, for example, less than about 5 seconds, less than about 2 seconds, less than about 1 second, less than about 0.5 second, or less than about 0.1 second.

[0079] The high-velocity jet **60** may produce significant mechanical power (e.g., tens of Watts in

some embodiments). When the jet **60** is directed toward the tooth **10**, a fraction of this mechanical power may go toward heating the tooth **10** as well as nearby teeth and gums. To avoid discomfort to the patient, in some embodiments, excess heat, if present, may be removed by, for example, irrigating the tooth under treatment with a stream of liquid (e.g., water at room temperature or cooler). The stream of liquid can absorb and carry away some or all of the excess heat, if present, that may be produced by the jet **60**. The stream of liquid may be aspirated from the patient's mouth. [0080] The methods described herein may be used as standalone treatments for root canal procedures, or they may be used in conjunction with other dental treatments (which may or may not involve liquid jet methods).

[0081] The high-velocity jet treatment methods described herein may be particularly effective in certain system operating ranges. For example, a jet having an energy flux greater than about 75 kW/cm² may be particularly effective.

[0082] The apparatus and methods described above may include additional devices and components configured to provide additional functionality to the endodontic treatment system. FIGS. **10A-10H** schematically illustrate embodiments of a contact member configured as a cap **120** that may be attached to and detached from the distal end **58** of the handpiece **50**. In some embodiments, the cap **120** may have threads that engage complimentary threads on the distal end **58** of the handpiece **50** (see, e.g., FIG. **10B**). The cap **120** can be fitted around the crown **12** of the tooth **10** and used to orient the jet **60** toward a suitable opening into the pulp chamber **26**. The cap **120** may be used to orient the distal end **58** of the handpiece **50** so that the liquid jet **60** is directed obliquely at a dentinal surface on the floor **82** (and/or sides) of the pulp cavity **26** and not directly down any of the canal space **30**. The cap **120** may be formed from a transparent or translucent material and may be sufficiently flexible to fit around teeth having a range of sizes. In some embodiments, the distal end **58** of the handpiece **50** may be rotatable and/or extendable with respect to the cap **120** so that the jet **60** may be moved closer to or further from a desired tooth portion. In some embodiments, a guide tube can extend distally and axially from the distal end **58** of the handpiece **50**, and provides a shielded pathway for the high velocity jet emanating from the orifice **66**. With regard to FIG. **12**, a guide tube **137** includes a distal opening aligned with the axis of the tube, and the interior bore of the tube is provided with a helical surface, similar to a rifling surface, that influences the passing jet beam to impart a turbulent, swirling effect thereto. This effect broadens the impact zone of the jet and reduces the cutting action, while augmenting other aspects of the jet effect, such as abrasion and emulsification. The cap **120** may include an outflow opening to permit organic material and liquid to be evacuated from the tooth **10**. Alternatively, the cap may include a suction port so that fluid may be removed from the pulp chamber at substantially the same volume as it is introduced. The suction port thus prevents any of the removed diseased tissue and liquid from entering the patient's mouth. Another aspect of the invention comprises introducing a volume of liquid in the form of a jet into a pulp chamber of a tooth and removing a volume of liquid from the pulp chamber at substantially the same rate that it is introduced. Preferably, removal is accomplished by suction. In certain embodiments, the distal end **58** of the handpiece **50** may include multiple orifices which provide multiple jets, and the cap **120** may be used to orient the handpiece **50** such that jets are directed not only at the floor **82** (and/or sides) of the pulp chamber **28**, but also towards entrances to the canal spaces **30**. In some embodiments, the handpiece **50** may also be tilted and rotated to allow the jet **60** to be aimed axially into all canal openings.

[0083] As schematically shown in FIGS. **10C-10H**, a plurality of caps **120** may be configured to fit over teeth of different sizes and shapes. Advantageously, each different cap **120** may be color coded to permit easy selection by the dentist during a treatment procedure. The cap **120** may also be sized and shaped to fit within an opening formed in the tooth **10** at the beginning of certain endodontic procedures (such as the openings **80**, **84**, and **88**; see FIGS. **6A-6C**), rather than fitting over or around the exterior of the tooth **10**.

[0084] As depicted in the partially exploded cross-section view shown in FIG. 10B, the distal end 58 of the handpiece 50 may include one or more pressure sensors 124. As the cap 120 is urged onto a tooth (or into an opening in the tooth), the pressure sensor 124 provides a signal that indicates when a sufficiently “tight” fit has been achieved. The pressure sensor 124 may be electrically connected to the controller 52, and enables the jet 60 to operate only when the sensor senses contact. A suitable audible, visible, and/or tactile signal may be output (e.g., by the user interface 54 or by an output device on the handpiece 50) to indicate that the cap 120 is in position on the tooth 10.

[0085] The system may also include a distance sensor to indicate the distance between the distal end 58 of the handpiece 50 and a surface of the tooth 10. The distance sensor may provide an audible, tactile, and/or visible indication when the distal end 58 is at a suitable distance from the surface for operation of the liquid jet 60 (e.g., not too close to damage the tooth and not too far for the jet to be ineffective). In one embodiment, the distance sensor comprises a pair of optical elements mounted on the handpiece and spaced from each other. Each optical element directs an optical beam that intersects the other beam a predetermined distance away from the handpiece 50. The operator of the handpiece 50 can maneuver the handpiece 50 until the intersecting beams illuminate a desired portion of the tooth 10 and then actuate the liquid jet 60. The optical elements may comprise light-emitting diodes (LEDs) and/or lasers. In some embodiments, more than two optical elements may be used (e.g., to indicate a range of distances).

[0086] A person of ordinary skill will recognize that a wide variety of sensors may be used in addition to or instead of the pressure sensor 124 and/or the distance sensor. For example, certain embodiments utilize one or more electric, magnetic, acoustic, and/or optical sensors to determine position and/or orientation of the handpiece 50 (or portions thereof) in the mouth. For example, proximity sensors, including capacitive sensors, ultrasonic sensors, light reflectance sensors, magnetic inductance sensors, and so forth, may be used. Certain embodiments may comprise one or more orientation sensors (e.g., accelerometers) configured to sense the orientation of the longitudinal jet axis 70 relative to one or more reference landmarks in the mouth (e.g., a portion of the tooth 10 such as the openings 80, 84, 88). The system may include a timer configured to deactivate the jet 60 after a predetermined time interval to reduce likelihood of damage to the tooth 10 and/or root canal system. Systems comprising one or more such sensors advantageously may provide increased safety. For example, certain such embodiments may prevent (or inhibit) an operator from actuating the liquid jet 60 until the distal end 58 of the handpiece 50 is suitably positioned and/or oriented adjacent the tooth 10.

[0087] As described herein, acoustic energy capable of producing cavitation may be particularly effective at cleaning the root canal system. It has been found that this acoustic energy may be efficiently produced by directing a high-velocity beam of liquid onto a portion of the tooth. However, the scope of the present disclosure is not limited to methods using high-velocity jets. In other embodiments, the acoustic energy is generated by vibrating mechanical devices (e.g., a piezoelectric transducer), ultrasonic (or megasonic) generators (e.g., an ultrasonic and/or a megasonic horn), or any other component capable of producing acoustic vibrations. FIG. 11 schematically illustrates one method for generating the acoustic wave 100 using a piezoelectric transducer 144. In this method, an enclosure 142 is attached to the tooth 10. The enclosure 142 comprises a chamber 140 filled with a liquid (e.g., water). The transducer 144 is disposed in or on the chamber 140. When actuated, the transducer 144 vibrates, which causes the acoustic wave 100 to propagate through the surrounding fluid and the tooth 10. The acoustic wave 100 cleans the root canal system substantially as described above. Because the vibrating transducer 144 is not in direct contact with the tooth 10, possible damage to the tooth 10 is reduced or eliminated. Although in FIG. 11 a mechanical transducer 144 is used to generate the acoustic wave 100, in other embodiments, the acoustic wave 100 may be produced by, for example, directing the high-velocity liquid jet into or onto the enclosure 142. In such embodiments, the enclosure 142 acts as an

“acoustic waveguide” converting jet kinetic energy into acoustic energy that propagates through the tooth **10** as the acoustic wave **100**. Beneficially, the liquid in the chamber **140** may absorb excess mechanical energy which may otherwise produce unwanted heat in the tooth **10**. In another embodiment, an ultrasonic horn is disposed near the enclosure **142** and used to generate the acoustic wave **100**.

[0088] Any of the procedures described herein may be carried out with the use of a rubber dam. Further although the tooth **10** depicted in the figures is a molar, one of ordinary skill in the art will appreciate that the procedures may be performed on any type of tooth such as an incisor, a canine, a bicuspid, or a molar. Also, the disclosed methods are capable of cleaning root canal spaces having a wide range of morphologies, including highly curved root canal spaces which are difficult to clean using conventional dental techniques. Moreover, the disclosed methods may be performed on human teeth (including children's teeth) and/or on animal teeth.

[0089] The foregoing description sets forth various preferred embodiments and other illustrative but non-limiting embodiments of the inventions disclosed herein. The description provides details regarding combinations, modes, and uses of the disclosed inventions. Other variations, combinations, modifications, equivalents, modes, uses, implementations, and/or applications of the disclosed features and aspects of the embodiments are also within the scope of this disclosure, including those that become apparent to those of skill in the art upon reading this specification. Additionally, certain objects and advantages of the inventions are described herein. It is to be understood that not necessarily all such objects or advantages may be achieved in any particular embodiment. Thus, for example, those skilled in the art will recognize that the inventions may be embodied or carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other objects or advantages as may be taught or suggested herein. Also, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence.

[0090] Accordingly, the scope of the inventions disclosed herein is to be determined according to the following claims and their equivalents.

Claims

1. (canceled)
2. A method for treating an infected or decaying condition in a tooth, the method comprising: substantially filling at least a portion of a treatment region of the tooth with a liquid medium; generating energy deliverable to a handpiece; directing an energy beam using the generated energy along a beam axis and into the liquid medium; and generating pressure waves with the energy beam in the liquid medium, the generated pressure waves having sufficient energy to remove at least some organic material from the treatment region of the tooth at which the energy beam is directed and at locations remote from the portion of the treatment region.
3. The method of claim 2, wherein directing the energy beam comprises directing a collimated and coherent energy beam along the beam axis and into the liquid medium.
4. The method of claim 3, wherein directing the energy beam comprises directing a liquid jet along the beam axis and into the liquid medium.
5. The method of claim 2, wherein substantially filling the at least the portion of the treatment region with the liquid medium comprises substantially filling at least a portion of a root canal.
6. The method of claim 2, wherein generating pressure waves comprises generating pressure waves having frequencies having a widespread range of acoustic noise.
7. The method of claim 6, wherein the frequencies comprise frequencies in a range from about 1 Hz to about 10 MHz.
8. The method of claim 2, further comprising removing the liquid medium from the tooth.

9. The method of claim 2, wherein directing the energy beam comprises orienting the handpiece to direct the energy beam along the beam axis.
10. The method of claim 9, wherein orienting the handpiece comprises positioning a cap on the treatment region of the tooth.
11. The method of claim 2, further comprising orienting an enclosure having a mounting surface against the treatment region of the tooth, wherein the enclosure is configured to define a chamber to be filled with the liquid medium when the enclosure is positioned against the tooth.
12. The method of claim 2, further comprising pulsing the energy beam.
13. The method of claim 2, wherein substantially filling the at least the portion of the treatment region with the liquid medium comprises substantially filling the treatment region with a substantially gas-free liquid.
14. The method of claim 2, further comprising removing substantially all organic material from one or more dental tubules with the generated pressure waves.
15. The method of claim 2, wherein directing the energy beam comprises generating an energy beam having an energy flux in a range of 1 kW/cm.² to 1000 kW/cm.².
16. The method of claim 15, wherein generating the energy beam comprises generating an energy beam having an energy flux in a range of 50 kW/cm.² to 750 kW/cm.².
17. The method of claim 2, wherein directing the energy beam comprises generating an energy beam having a transverse cross-section in a direction perpendicular to the beam axis, the transverse cross-section having a center surrounded by a peripheral region, wherein a power of the energy beam is larger at the center than at the peripheral region.
18. The method of claim 17, wherein the center lies in a central region surrounded by the peripheral region, and wherein the power of the generated beam is constant across the central region.
19. The method of claim 2, wherein directing the energy beam comprises directing the energy beam through an aperture.
20. The method of claim 2, wherein generating energy deliverable to a handpiece comprises activating a compressor system.)
21. The method of claim 2, wherein directing the energy beam comprises directing a collimated energy beam along the beam axis and into the liquid medium.
22. The method of claim 21, wherein an angular divergence of the collimated energy beam is less than about 1 degree.
23. A method for treating an infected or decaying condition in a tooth, the method comprising: substantially filling at least a portion of a treatment region of the tooth that includes the infected or decaying condition with a liquid medium, the treatment region including at least a root canal and one or more dentinal tubules extending from the root canal; generating energy deliverable to a handpiece; directing an energy beam using the generated energy along a beam axis and into the liquid medium present at the portion of the treatment region of the tooth; and generating pressure waves within the liquid medium having sufficient energy to remove at least some organic material from the tooth at the portion of the treatment region at which the energy beam is directed and the one or more dentinal tubules.
24. The method of claim 23, wherein directing the energy beam comprises directing a collimated and coherent energy beam along the beam axis and into the liquid medium.
25. The method of claim 24, wherein directing the energy beam comprises directing a liquid jet along the beam axis and into the liquid medium.
26. The method of claim 23, wherein substantially filling at least the portion of the treatment region of the tooth that includes the infected or decaying condition with the liquid medium comprises substantially filling at least the portion of the treatment region of the tooth that includes the infected or decaying condition with a substantially gas-free liquid.
27. The method of claim 23, wherein generating pressure waves comprises generating pressure waves having frequencies having a widespread range of acoustic noise.

- 28.** The method of claim 27, wherein the frequencies comprise frequencies in a range from about 1 Hz to about 10 MHz.
- 29.** The method of claim 23, wherein directing the energy beam comprises generating an energy beam having an energy flux in a range of 1 kW/cm.² to 1000 kW/cm.².
- 30.** The method of claim 22, wherein an angular divergence of the energy beam is less than about 1 degree.
- 31.** A method for treating an infected or decaying condition in a root canal of a tooth, the method comprising: providing a liquid medium to the tooth in the root canal and in remote locations; generating energy configured to create an energy beam; directing the energy beam along a beam axis and into the liquid medium without requiring that the energy beam be directed into the remote locations; and generating pressure waves within the liquid medium having sufficient energy to remove at least some organic material from the root canal and the remote locations, the remote locations including a plurality of dentinal tubules extending from the root canal.
- 32.** The method of claim 31, wherein directing the energy beam comprises directing a collimated and coherent energy beam along the beam axis and into the liquid medium.
- 33.** The method of claim 31, wherein directing the energy beam comprises directing a liquid jet along the beam axis and into the liquid medium.
- 34.** The method of claim 31, further comprising removing the liquid medium from the tooth.
- 35.** The method of claim 31, wherein generating energy comprises activating a compressor system.
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